

Data — CRSP Daily, 2024-01-02 to 2024-12-31

Actual rows at the beginning and end of the sample; one row is one security-day.

date	PERMNO	openprc	prc	vol	ret	shrout
01-02	10026	165.97	168.86	89,969	0.010291	19,367
01-02	10028	4.88	4.71	18,599	-0.030864	26,700
01-02	10032	106.36	106.35	78,780	-0.016462	27,504
01-02	10044	4.69	4.94	7,736	0.073913	6,304
01-02	10065	17.60	17.45	235,208	-0.014681	120,810
01-02	10066	3.39	3.35	32,753	-0.011800	11,784
12-31	10026	156.23	155.13	52,207	-0.001095	19,478
12-31	10028	7.22	7.18	9,659	-0.001391	25,996
12-31	10032	157.50	156.48	101,562	-0.001340	27,092
12-31	10044	2.40	2.43	10,438	-0.004098	7,598
12-31	10065	20.36	20.20	219,627	-0.004926	112,690
12-31	10066	4.88	4.90	24,310	0.006160	11,784

Source: data/crsp_dsf_2024.csv.gz

Data — Point-in-time filter build

Security-day stage	Rows
Raw CRSP Daily	2,400,962
Effective-date name match	2,400,962
Common shares and main exchanges	987,277
Signal-day $ \text{prc} \geq \5	702,841
Complete features and available label horizon	633,995

$\text{SHRCD} \in \{10, 11\}$ and $\text{EXCHCD} \in \{1, 2, 3\}$ are evaluated on each signal date. Raw file: 252 trading days and 10,503 PERMNOs.

Source: `data_audit.csv`; `run_metadata.json`

Data — Check the meanings and units

RET is a holding-period return

For PERMNO 10026 on Jan 2,
openprc = 165.97, prc = 168.86, and
ret = 0.010291.

$$168.86/165.97 - 1 = 0.01741 \neq 0.010291.$$

The open-to-close price change is a different quantity.

Units used in features

$$\text{shares} = 1000 \text{ shROUT}$$

$$\text{market cap} = |\text{prc}| \times \text{shares}$$

CRSP negative prc marks a bid/ask average; |prc| supplies price magnitude.

Source: data/crsp_dsf_2024.csv.gz; WRDS CRSP data definitions

Data — Eight predictors from the trailing 20 market days

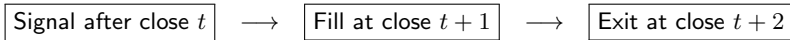
Let W_t be the 20-market-day window ending at signal close t . Each statistic requires at least 10 observations.

Feature	Definition on W_t
Trading days	Count of valid-price observations
Median price	Median $ prc $
ADV	Mean share volume
Median dollar volume	Median $ prc \times vol$
ADV dollar	Mean $ prc \times vol$
Median turnover	Median $vol/(1000 \text{ shrout})$
Annualized volatility	$\sqrt{252} \text{ sd}(\text{ret})$
Median market cap	Median $ prc \times 1000 \text{ shrout}$

Source: `src/run_crsp_benchmark_2024.py`

Data — Response and executable timing

i denotes PERMNO and t denotes a market day. $X_{i,t} \in \mathbb{R}^8$ is known after close t .



Primary response: realized holding-period return

$$Y_{i,t} = R_{i,(t+1,t+2]} = \begin{cases} (1+r)(1+r^{DL}) - 1, & r, r^{DL} \text{ both observed,} \\ r \text{ or } r^{DL}, & \text{only one observed.} \end{cases}$$

Auxiliary direction label: $Z_{i,t} = \mathbf{1}\{Y_{i,t} > 0\}$. No label is formed if the entry cannot fill or both return fields are missing.

Source: `doc/pipeline.md`; `src/run_crsp_benchmark_2024.py`

Problem — Estimate return and daily cross-sectional rank

Prediction in fold k with model m

$$\hat{Y}_{i,t}^{(m,k)} = f_{m,k}(X_{i,t}), \quad f_{m,k} : \mathbb{R}^8 \longrightarrow \mathbb{R}.$$

$f_{m,k}$ is fitted using only the fold's training and validation data; $\hat{Y}$ is its return forecast.

$$\text{MSE}_m = \frac{1}{N} \sum_{i,t} (Y_{i,t} - \hat{Y}_{i,t}^{(m,k)})^2, \quad \text{RankIC}_{m,t} = \rho_S \left(\hat{Y}_{\cdot,t}^{(m,k)}, Y_{\cdot,t} \right).$$

ρ_S is Spearman correlation among stocks observed on day t . Secondary test: estimate $P(Z_{i,t} = 1 \mid X_{i,t})$. Economic test: rank-based long-short return after costs.

Source: `doc/pipeline.md`; `src/run_crsp_monthly_walkforward_2024.py`

Approach — Daily observations, monthly walk-forward

Test block	Train through	Validate	Test dates	Train rows	Val. rows	Test rows
July	May 29	Jun 3–26	Jul 1–31	233,220	46,591	60,347
August	Jun 26	Jul 1–29	Aug 1–30	285,313	54,804	59,828
September	Jul 29	Aug 1–28	Sep 3–30	345,555	54,349	54,400
October	Aug 28	Sep 3–26	Oct 1–31	405,407	48,927	62,431
November	Sep 26	Oct 1–29	Nov 1–26	459,792	56,950	49,040

Expanding train



Prior-month validation



Daily test forecasts

Two-market-day label gap at each boundary; December outcomes stay sealed.

Source: `run_metadata.json`

Approach — Fixed baselines and fitted models

Model	Forecast or fitting rule
Zero	$\hat{Y} = 0$; MSE reference
Reversal	$\hat{Y} = -\text{ret}_{i,t}$; fixed one-day reversal score
Ridge	$\min_{\beta} \ Y - X\beta\ _2^2 + \alpha\ \beta\ _2^2$; $\alpha \in \{0.1, 10, 1000\}$
Boosted tree	Histogram gradient boosting; 80 rounds, learning rate 0.05, leaves $\in \{3, 7\}$
Logistic	$P(Z = 1 X) = \sigma(\beta_0 + X^\top \beta)$; $C = 1$

Ridge and tree settings minimize prior-month validation MSE. Final regression fit uses train + validation; median imputation and scaling are fitted within each fold.

Source: `src/run_crsp_monthly_walkforward_2024.py`

Approach — Rank portfolio and cost equation

On each signal day, select top and bottom predicted-return deciles; L_t and S_t are the selected sets.

$$w_{i,t} = \begin{cases} +1/|L_t|, & i \in L_t, \\ -1/|S_t|, & i \in S_t, \\ 0, & \text{otherwise.} \end{cases}$$

$$r_t^{\text{gross}} = \sum_i w_{i,t} Y_{i,t}, \quad \text{TO}_t = \sum_i |w_{i,t} - w_{i,t-1}|, \quad r_t^{\text{net}} = r_t^{\text{gross}} - \underbrace{10^{-3}}_{10 \text{ bps per turnover unit}} \text{TO}_t.$$

Both sides are equal weight. A portfolio date is scored only if every selected holding has an observed outcome.

Source: `src/run_crsp_benchmark_2024.py`; `run_metadata.json`

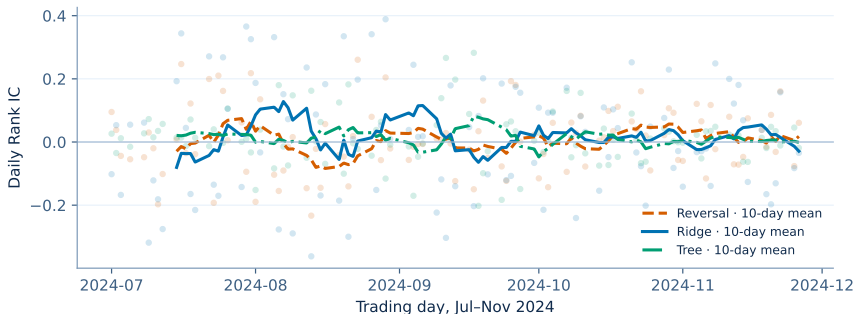
Result — Prediction gains across 105 daily tests

Model	Test MSE	Mean IC	Mean Rank IC
Zero	0.00131959	—	—
Reversal	0.00267757	0.00316	0.00590
Ridge	0.00131761	0.00666	0.01104
Boosted tree	0.00131746	0.01364	0.01276

285,882 of 286,046 daily test predictions have valid labels. The tree leads in Rank IC; its MSE gain relative to zero is about 0.16%.

Source: `model_comparison.csv`; `predictions.csv.gz`

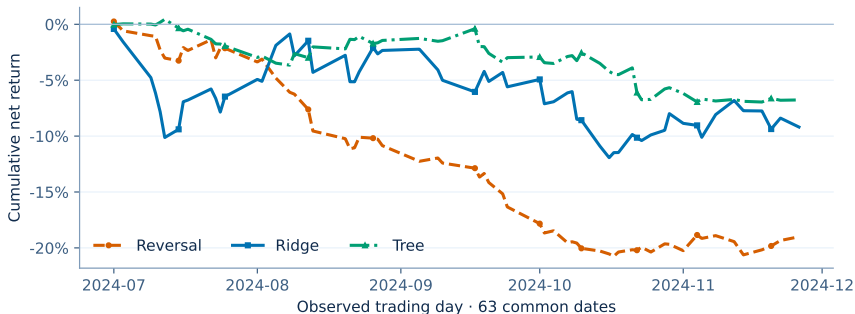
Result — Daily cross-sectional Rank IC



Dots represent individual trading days; lines are 10-day rolling means. Color and line pattern distinguish reversal, Ridge, and tree.

Source: `ic_reversal.csv`; `ic_ridge.csv`; `ic_tree.csv`

Result — Net portfolio return on observed days



Cumulative net return uses 63 dates with complete selected-holding outcomes for all three strategies. It is a conditional comparison.

Source: `portfolio_reversal.csv`; `portfolio_ridge.csv`; `portfolio_tree.csv`

Result — Trading costs and direction classification

Portfolio	Net Sharpe	Mean turnover
Reversal	-6.350	3.31
Ridge	-1.527	0.33
Boosted tree	-2.852	0.20

Logistic classifier

ROC-AUC: 0.5032

Balanced accuracy: 50.41%

Cross-entropy: 0.69293

Sharpe uses 63 common dates at 10 bps per turnover unit. Selected-holding outcomes are incomplete on 15, 36, and 21 of the 105 test days for reversal, Ridge, and tree.

Source: `model_comparison.csv`; `classification_metrics.csv`

Conclusion — Small ranking signal, limited trading evidence

1. Boosted tree has the highest stitched Rank IC (0.0128), a small edge over Ridge (0.0110).
2. Direction classification is near chance (AUC 0.5032).
3. All three portfolios have negative net Sharpe on the 63 comparable dates under the stated cost.
4. Missing selected-holding outcomes limit inference about the complete 105-day portfolio path.

Next step

Resolve missing exit returns, then rerun the same daily portfolio protocol on a complete set of test dates.

Source: `model_comparison.csv`; `classification_metrics.csv`; `run_metadata.json`